

PrivaLens

Automated Web Privacy & Regulatory Compliance Scanner

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NLP Policy Engine & Verification

What is PrivaLens?

An automated web platform where users enter any website URL. Headless Puppeteer navigates the site, intercepts all live HTTP/WebSocket requests, identifies hidden trackers and PII leaks, and cross-checks actual data flows against the site's stated Privacy Policy using NLP.

Continuous Evaluation Objective

Demonstrates complete software engineering execution: asynchronous distributed job queues (Bull/Redis) to handle non-blocking browser scraping, decoupled Python NLP microservice for legal text parsing, and structured PostgreSQL storage.

Motivation: Why Build PrivaLens?

Addressing modern privacy regulations, technical challenges, and software architecture.

1. Stricter Privacy Regulations

Urgent Industry Need

- Laws like EU GDPR, India's DPDP Act 2023, and CCPA impose massive penalties for unconsented data tracking.
- Small businesses & startups cannot afford expensive manual legal audits (\$5k–\$15k per audit).
- Provides an automated self-serve tool that developers can run before deploying new code.

2. Novel Multi-Tech Integration

Scraping + NLP + Legal Tech

- Combines headless browser automation with natural language understanding of legal privacy policies.
- Goes beyond static code linters by auditing real runtime HTTP packets and dynamic third-party scripts.
- Highly practical project demonstrating real-world cybersecurity and data engineering skills.

3. Asynchronous Architecture

Distributed Systems Practice

- Crawling dynamic web pages and sniffing network traffic takes time (15–60 seconds per site).
- Implements decoupled worker queues (Bull/Redis) to prevent UI blocking during heavy scans.
- Direct application of core software engineering and microservices principles.

Problem Statement: The Web Privacy Compliance Gap

Modern web applications frequently leak user data without the developers or owners realizing it.

1. Hidden PII Leaks & Trackers

Unmonitored runtime data leaks

- Third-Party Scripts: Websites embed analytics, chat widgets, and ad tags that quietly harvest user data.
- Query Parameter Leaks: Personal data (emails, usernames, IPs) often gets leaked in plain HTTP GET parameters.
- Unauthorized Cookies: Tracking cookies are often stored before users give explicit consent.

2. Policy vs. Reality Discrepancy

Privacy policy contradicts live code

- False Claims: Policies often state 'We do not share your personal data with any third parties'.
- Live Network Reality: Runtime logs show data packets firing directly to external ad brokers & analytics endpoints.
- Severe Legal Risk: Direct violation of DPDP Act and GDPR deceptive trade rules.

3. Lack of Accessible Audit Tools

High cost and static tool limitations

- Manual Audits are Prohibitive: Security firms charge thousands of dollars, making regular checks impossible.
- Static Scanners Miss Runtime Calls: Standard linters only read code files and cannot track dynamic network beacons.
- Need for Automated Testing: Developers lack an automated way to verify compliance on demand.

Proposed Solution: PrivaLens System Architecture

A 3-tier architecture combining headless crawling, asynchronous job queues, and NLP policy parsing.

Frontend Dashboard

React / Next.js

- Simple URL input box with live WebSocket scan progress.
- Interactive compliance scorecard with risk levels (High / Med / Low).
- Visual table of all detected trackers, cookies, and leaked fields.
- Downloadable PDF report with actionable developer fixes.

Headless Engine & Queue

Node.js + Puppeteer + Redis

- Puppeteer Headless Crawler: Loads the page, executes JS, and logs every outgoing network request.
- Payload Inspector: Checks headers, query parameters, and cookie storage for PII.
- Bull/Redis Queue: Runs crawls asynchronously in the background.

NLP Engine & Storage

Python / NLP + PostgreSQL

- Policy Scraper & NLP Parser: Extracts data-handling statements from the website's Privacy Policy page.
- Discrepancy Matrix: Compares NLP claims against detected network trackers.
- PostgreSQL: Saves scan history and compliance metrics.

PROJECT ROADMAP & CONTINUOUS EVALUATION PHASES

Phase 1: Puppeteer Network Interceptor → Phase 2: Bull/Redis Queue & Postgres → Phase 3: Python NLP Policy Matcher → Phase 4: Next.js Scorecard UI